Probing Cross-Lingual Lexical Representations in Multilingual Large Language Models for Low-Resource Dyslexia Screening

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Please change your title according your work

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Approval

The thesis titled "Probing Cross-Lingual Lexical Representations in Multilingual Large Language Models for Low-Resource Dyslexia Screening" submitted by

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1.5 Methodology in Brief

Four publicly released multilingual LLMs are loaded in their official pretrained form and frozen. The 440-stimulus dataset is fed through each model with a single forward pass per item; hidden states are collected at every layer (or every encoder layer for the encoder-decoder model). For each phonological task, a per-layer linear probe (logistic regression for classification, ridge regression for syllable count) is trained on English and evaluated on English (in-language) and on Bangla (zero-shot cross-lingual). Surface-feature baselines are trained on the same labels using only character statistics. After length matching, the surviving task is re-probed and used to score Representational Similarity Analysis (RSA) and a forward-hook causal ablation that zeroes the top-K probe-ranked hidden units and remeasures transfer. The mT5 encoder is re-ablated at five depths with random-unit controls to test whether localisation is layer-specific or persistent. Figure 4.1 (Chapter 4) summarises the pipeline.

1.6 Scopes and Challenges

The scope is bounded to (i) four multilingual LLM families released as of 2024, (ii) two languages (English and Bangla), (iii) word-level stimuli with sentence-level items reserved for descriptive use, and (iv) frozen, read-only probing – no fine-tuning, no continual pretraining and no clinical deployment. The four main challenges encountered were: (1) 8B-parameter causal models do not fit in a single consumer GPU with full-precision activations across 32 layers; this is mitigated with 4-bit weight quantisation via Unsloth/BitsAndBytes [16] for the Llama and BLOOM passes, while keeping probes trained on the resulting hidden states. (2) Bangla orthography includes conjuncts (*yuktoborno*) whose tokenisation varies across model families, biasing surface metrics; this is addressed by computing length statistics from the original Unicode string, not the tokeniser output. (3) Most Bangla phonological labels are not publicly available; we generate them from phonotactic rules and verify with a native speaker, but discover during sanity checking that the voicing label set is fully recoverable from the first character – a finding that reframes part of the original research question. (4) Causal ablation specificity requires a random-unit control with multiple seeds; the encoder-decoder localisation analysis is therefore the most compute-intensive part of the pipeline and is run on the smaller mT5 encoder.

1.7 Thesis's organization

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Table 3.1: One-year thesis timeline. Phases overlap at their boundaries; writing runs in parallel with the final analysis phases.

Phase	Timeline	Key activities
Phase 1	Months 1–2	Literature review, problem formulation and definition of the four research questions.
Phase 2	Months 2–3	Stimulus design and native-speaker verification of the 440-item English/Bangla dataset.
Phase 3	Months 3–5	Representation extraction across the four model families (frozen forward passes, per-layer hidden-state caching).
Phase 4	Months 5–7	Per-layer probing across the four tasks; English-only and cross-lingual evaluation; first result tables and plots.
Phase 5	Months 7–8	Surface-feature sanity checks; reframing of three of the four tasks; length-matched stimulus reconstruction and re-probing.
Phase 6	Months 8–10	Representational Similarity Analysis, causal forward-hook ablation, and the mT5 depth-and-overlap analysis.
Phase 7	Months 10–12	Thesis writing, figure preparation, internal review and defence preparation.

Resources: one supervisor (Dr. Farig Yousuf Sadeque), one co-supervisor (Mr. Md. Tanzim Reza), four student investigators, one consumer GPU, and one native Bangla reviewer (for stimulus verification, ~4 hours).

Gantt chart ?

Budget

Because the methodology relies on frozen on-device probing rather than model training or paid cloud APIs, the marginal compute cost of the study is negligible (BDT 120 in electricity; see the Economic Analysis in Section 3.8). The remaining budget is dominated by student-side overheads typical of a one-year thesis. Table 3.2 itemises the estimated total budget in Bangladeshi Taka (BDT).

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4.2 Preliminary Design or Design (Model) Specification

Three design alternatives were considered before settling on frozen per-layer linear probing.

Alternative A: Fine-tune each model on each task. Highest expected performance, but conflates model knowledge with task-specific adaptation, requires large labelled corpora that do not exist for Bangla phonology, and is roughly 10^6 times more expensive in compute than frozen probing. Rejected.

Alternative B: Single-layer (final hidden state) probing. Standard practice in some sentence-level probing studies, but discards information about *where* a feature emerges in the network and prevents the layer-emergence comparison plot (Figure 5.5). Rejected.

Alternative C (selected): Per-layer linear probing with surface-feature baselines and causal ablation. Frozen models, linear probes (logistic regression for classification, ridge for regression), per-layer evaluation, length-matched stimulus subsets, RSA and forward-hook ablation. This alternative isolates representational content from model behaviour, exposes layer-wise emergence, and supports causal-localisation claims via random-unit controls. Selected.

4.2.1 Models and architectural specification

Table 4.1 summarises the four probed models.

Table 4.1: Architectural specification of the four probed multilingual LLMs.

Model	Architecture	Tokeniser	Layers	Hidden	Bangla coverage
XLM-R large	encoder-only	SentencePiece	24	1024	explicit
Llama 3.1 8B	decoder-only	tiktoken BPE	32	4096	minimal
BLOOM 7B1	decoder-only	BPE	30	4096	explicit (ROOTS)
mT5 XL	encoder-decoder	SentencePiece	24 (encoder)	2048	explicit

For mT5 XL only the encoder is probed; the decoder is held frozen and not used for representation extraction. Figure 4.2 compares the four architectures side by side.

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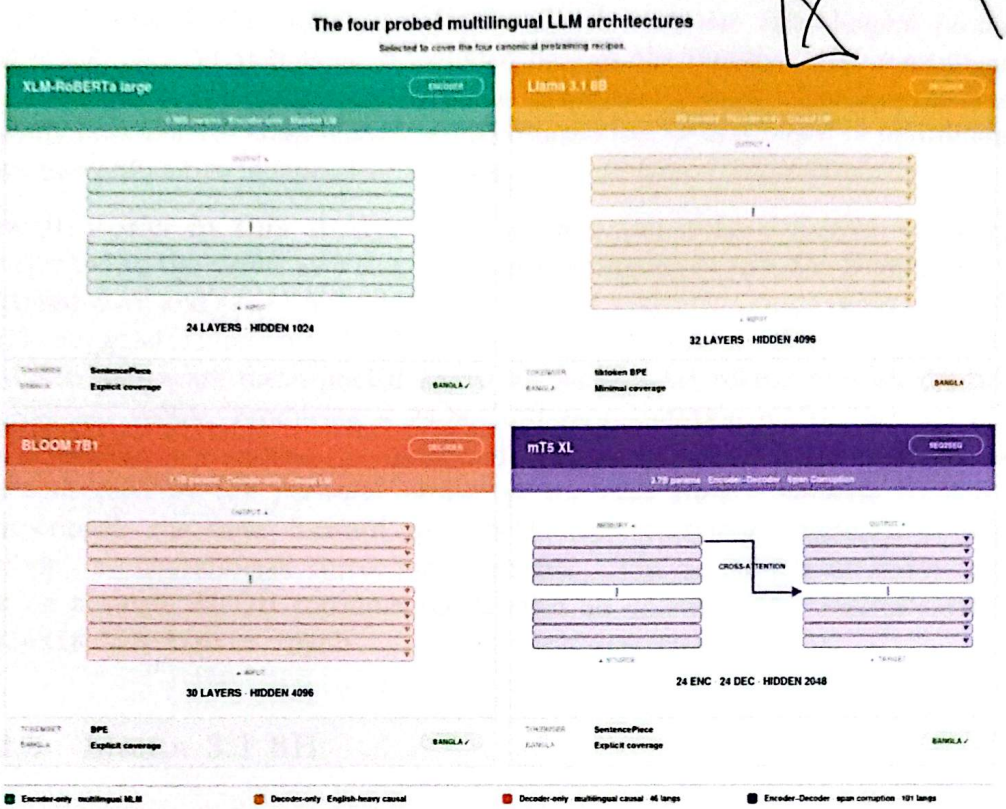


Figure 4.2: Side-by-side comparison of the four probed multilingual LLM architectures. XLM-R large is an encoder-only multilingual masked language model; Llama 3.1 8B is an English-heavy causal decoder; BLOOM 7B1 is a 46-language causal decoder with explicit Bangla coverage; mT5 XL is a 101-language span-corruption encoder-decoder. Layer count, hidden size, tokeniser and Bangla coverage are annotated per column.

The four models were selected to exhaust the four canonical pretraining recipes in multilingual NLP as of 2024: a multilingual masked encoder, an English-heavy causal decoder, a multilingual causal decoder, and a multilingual span-corruption encoder-decoder. The following subsections describe each model individually, focusing on the design choices that are most relevant to cross-lingual lexical probing.

4.2.2 XLM-RoBERTa large

XLM-RoBERTa large [10] is a 0.56-billion-parameter encoder-only Transformer trained with masked language modelling (MLM) on a 100-language version of the CommonCrawl corpus (CC-100). The model uses a 24-layer encoder stack with hidden dimension 1024 and a SentencePiece tokeniser with a 250,000-token vocabulary built jointly across all 100 pretraining languages. Bangla is present in the pretraining mix as an explicitly covered language, and SentencePiece allows Bangla conjunct characters (*yuktoborno*) to be tokenised into subword units rather than discarded. The MLM objective forces the encoder to learn bidirectional contextual representations that integrate information from both left and right context at every

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extraction. A single forward pass yields the 24 encoder-layer hidden states; per-layer mean pooling over the input tokens produces a 24-by-2048 representation matrix per stimulus, cached to `dyslexia-mt5/representations/`. The encoder is bidirectional like XLM-R's, so no causal-mask correction is needed. Because the encoder is only 1.67B parameters, mT5 XL fits in 16-bit precision on a single 24 GB consumer GPU without quantisation, which makes mT5 the most expensive but also the highest-fidelity model in the study and is the reason the depth-scan ablation (five `measure_L` values, three random-unit seeds each) is run on mT5 specifically and not on the larger BLOOM and Llama decoders. Implementation: `dyslexia-mt5/main.ipynb`.

4.2.6 Why these four

Each of the four models is the strongest publicly released representative of its pretraining recipe that runs on a single consumer GPU under 4-bit quantisation. Together they span the two-by-two design grid that defines the multilingual LLM landscape in 2024: {*encoder-only, decoder-only, encoder-decoder*} crossed with {*multilingual, English-heavy*}. Comparing them on the same surface-controlled stimuli with the same probes is the minimal experimental design that lets the thesis make claims about which pretraining recipe carries cross-lingual lexical structure. The four-model comparison is also a precondition for the model-selection recommendation in Chapter 6: without all four families on the same evaluation, it would not be possible to argue that BLOOM is the best practical substrate for a low-resource Bangla screening prototype and that mT5 is the best interpretability substrate.

4.3 Data Collection

4.3.1 Stimulus dataset

The dataset is a single CSV (`dyslex_probe_stimuli_clean.csv`, 440 rows, 18 columns) committed to each per-model repository. It covers 220 English and 220 Bangla items across seven task categories: minimal pair (60 per language), multisyllabic (30), pseudoword (25), real word (25), phoneme-grapheme (25), rhyme (20) and sentence (15). The detailed per-category counts and the eighteen-column schema are reproduced in Appendix B.

The Bangla side of the stimulus design is grounded in the Master's dissertation work of Badsha [9], who documented the phonological difficulty patterns of five dyslexic Bangla-speaking primary-school children at the University of Dhaka and published the consolidated findings in IJFMR (vol. 6, no. 3, May-June 2024). Three elements of our stimulus design are taken directly from that work: (i) the five voiced/unvoiced stop pairs (ক/গ, প/ব, ত/দ, ট/ড, চ/জ) used as the Bangla minimal-pair category, (ii) the multisyllabic-word and *matra* error patterns that informed the multisyllabic category, and (iii) the human difficulty ranking (voicing hardest to syllable-count easiest) against which we later compare the four-model ablation pattern in Chapter 5.

Bangla items are stored in Unicode Bengali script with author-assigned IPA, syllable

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macro F1 for classification and R^2 for regression, both averaged over a 5-fold cross-validation within the English training split for layer-selection robustness. Figure 4.3 illustrates the per-layer probing setup.

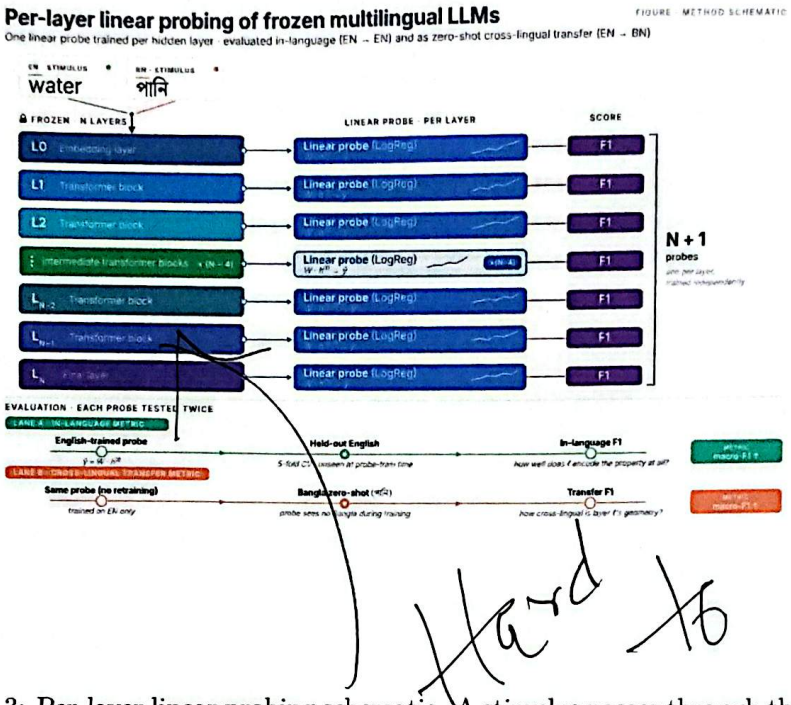


Figure 4.3: Per-layer linear probing schematic. A stimulus passes through the frozen model; each layer's mean-pooled hidden state feeds a separately trained linear probe. The English-trained probe is then evaluated on held-out English (in-language) and on the full Bangla subset (zero-shot cross-lingual transfer).

4.4.3 Surface-feature sanity checks

Three sanity-check baselines are run.

First-character lookup (voicing). For each Bangla voicing label, the leave-one-out probe is trained on the layer-0 (token-embedding) hidden state with a one-hot encoding of the first character. The resulting F1 quantifies how much voicing information is recoverable from the first character alone (Section 5.1.3, Figure 5.6).

Five-feature character-statistic baseline (ortho, syllable). A logistic / ridge baseline is trained on five hand-crafted features per item: character length, byte length, unique-character count, vowel ratio, longest consonant run. The baseline is run separately for English and Bangla. Any LLM probe whose F1 / R^2 does not exceed this baseline by more than 0.03 is interpreted as not contributing information above surface statistics.

Length matching (realword). Real and pseudo word items are matched 1:1 on character length within each language. Probes trained on the matched subset are reported as the only surface-controlled realword numbers.

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4.4.6 mT5 depth-scan and unit-overlap analysis

The mT5 encoder ablation is repeated at five depths (`measure_L` $\in \{4, 8, 12, 16, 20\}$) with three random seeds per depth. At each depth the top-10 probe-ranked units are recorded; the pairwise Jaccard overlap of the top-10 sets across depths is then tabulated. A unit that appears in the top-10 at three or more depths is reported as a *persistent backbone* unit; in the mT5 encoder these are u782 and u1880. The analysis answers two questions: (i) does localisation persist across depth or is it specific to one layer; (ii) are the same units carrying the information across depths or does each depth use a layer-local set. Figure 4.5 previews the expected hybrid pattern that the depth-scan ultimately confirms in Chapter 5.

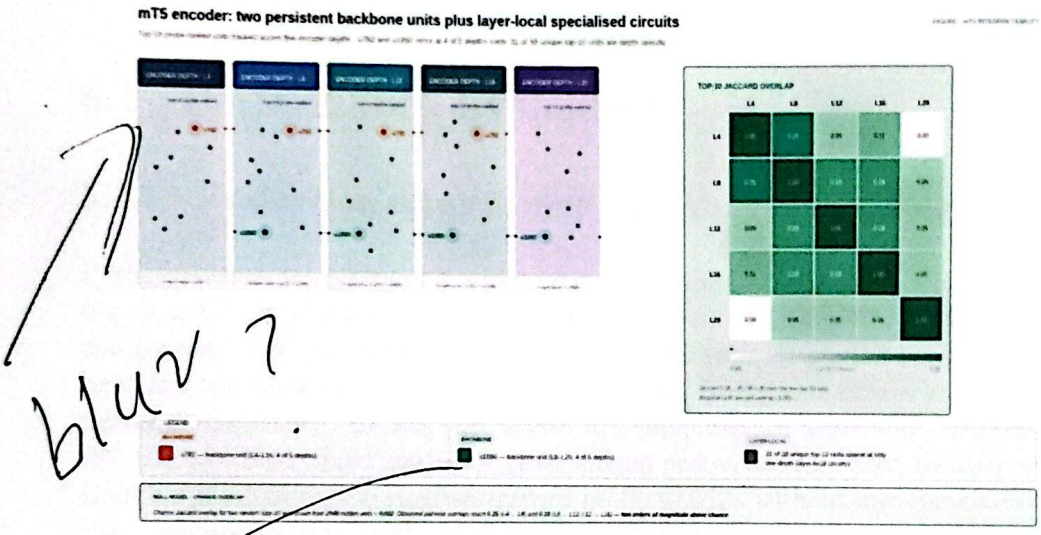


Figure 4.5: Schematic of the hybrid backbone-plus-routing pattern probed in the mT5 encoder. Two units (u782, u1880) appear in the top-10 probe-ranked set across four of the five probed encoder depths and constitute a persistent backbone; the remaining top-10 units at each depth are layer-local specialised circuits.

4.4.7 Reproducibility

Every result table in Chapter 5 corresponds to a deterministic CSV in the per-model results directory: `results_english_probing_*.csv`, `results_transfer_*.csv`, `results_rsa_*.csv`, `results_ablation_*.csv`, `results_*_realword_matched.csv` and (for mT5) `results_mt5_localization_depth_scan.csv` and `results_mt5_unit_overlap.csv`. Random seeds, scikit-learn versions and CUDA versions are pinned in Appendix A. The four-model integration is emitted to `comparison_four_models.md` and `comparison_four_models.xlsx` in `dyslexia-mt5/`.

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Real vs. pseudo word: survives length matching

When real and pseudo words are matched 1:1 on character length (50 matched per language; mean character length 6.88 in English, 5.20 in Bangla), all four models gain at least +0.20 F1 over both their layer-0 and the five-feature surface baseline on Bangla (Table 5.3). The real-versus-pseudo word task is the only one of the four that passes the surface-feature controls and is the only task on which the remainder of this chapter reports cross-lingual transfer, RSA and causal-ablation numbers.

Table 5.3: Length-matched realword probing per language. EN/BN at layer 0 versus peak layer, with gain over L0. All four models gain $\geq +0.20$ F1 on Bangla; this is the only task that passes surface-feature controls.

Model	EN L0	EN peak	EN gain	BN L0	BN peak	BN gain
XLM-R	0.81	0.88	+0.06	0.66	**0.92**	**+0.25**
Llama	0.75	0.96	+0.21	0.64	**0.90**	**+0.26**
BLOOM	0.81	0.94	+0.13	0.64	**0.94**	**+0.30**
mT5	0.65	0.90	+0.23	0.72	**0.92**	**+0.20**

5.1.4 RQ2: Length-matched cross-lingual transfer

The English-trained length-matched realword probe is evaluated zero-shot on the Bangla length-matched subset. Table 5.4 reports the surface-controlled cross-lingual numbers; Figures 5.7 and 5.8 visualise the per-layer trajectories and the headline surface-controlled comparison respectively.

Table 5.4: Length-matched cross-lingual transfer (EN $\rightarrow$ BN). Peak layer / peak BN F1 / gain over surface baseline / verdict. After surface controls only XLM-R and BLOOM transfer; Llama and mT5 collapse.

Model	Peak L (rel.)	Peak BN F1	Gain over surface	Verdict
XLM-R	L16 / rel. 0.67	**0.86**	**+0.24**	real transfer
BLOOM	L25 / rel. 0.83	**0.92**	**+0.30**	real transfer
Llama	L31 / rel. 0.97	0.63	+0.01	collapsed to surface
mT5	L20 / rel. 0.83	0.60	−0.02	collapsed below surface

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